

```

# 5 x 4 matrix

values <- as.numeric(strsplit(
  readline("Enter 20 values for 5x4 matrix: "), " ")
)[[1]])

m1 <- matrix(values, nrow = 5, ncol = 4, byrow = TRUE)

rownames(m1) <- paste0("R", 1:5)
colnames(m1) <- paste0("C", 1:4)

cat("\n5 x 4 Matrix:\n")

print(m1)

# 3 x 3 matrix

values <- as.numeric(strsplit(
  readline("Enter 9 values for 3x3 matrix: "), " ")
)[[1]])

m2 <- matrix(values, nrow = 3, ncol = 3, byrow = FALSE)

rownames(m2) <- paste0("R", 1:3)
colnames(m2) <- paste0("C", 1:3)

cat("\n3 x 3 Matrix:\n")

print(m2)

# 2 x 2 matrix

values <- as.numeric(strsplit(
  readline("Enter 4 values for 2x2 matrix: "), " ")
)[[1]])

m3 <- matrix(values, nrow = 2, ncol = 2, byrow = TRUE)

rownames(m3) <- c("Row1", "Row2")
colnames(m3) <- c("Col1", "Col2")

cat("\n2 x 2 Matrix:\n")

```

```
print(m3)
```

The screenshot displays the RStudio environment with the following components:

- Source Editor:** Contains R code for creating matrices `m2` and `m3`, and a loop for user input.
- Console:** Shows the execution of the code, including prompts for matrix dimensions and values, and the resulting matrix structures.
- Environment Pane:** Lists the objects in the global environment: `m1`, `m2`, `m3`, and `values`.

Source Editor Code:

```
19  
20 m2 <- matrix(values, nrow = 3, ncol = 3, byrow = FALSE)  
21  
22 rownames(m2) <- paste0("R", 1:3)  
23 colnames(m2) <- paste0("C", 1:3)  
24  
25 cat("\n3 x 3 Matrix:\n")  
26 print(m2)  
27  
28  
29 # 2 x 2 matrix  
30 values <- as.numeric(strsplit(  
31   readline("Enter 4 values for 2x2 matrix: "), " "  
32   )[[1]])  
33  
34 m3 <- matrix(values, nrow = 2, ncol = 2, byrow = TRUE)  
35  
36 rownames(m3) <- c("Row1", "Row2")  
37 colnames(m3) <- c("Col1", "Col2")  
38  
39 cat("\n2 x 2 Matrix:\n")  
40 print(m3)  
328 (Top Level) :
```

Console Output:

```
> source("C:/Users/2021e/Desktop/EX 1.R")  
Enter 20 values for 5x4 matrix: 1 2 3 4 5 6 7 8 9 10 11 12 13  
14 15 16 17 18 19 20  
  
5 x 4 Matrix:  
C1 C2 C3 C4  
R1 1 2 3 4  
R2 5 6 7 8  
R3 9 10 11 12  
R4 13 14 15 16  
R5 17 18 19 20  
Enter 9 values for 3x3 matrix: 1 2 3 4 5 6 7 8 9  
  
3 x 3 Matrix:  
C1 C2 C3  
R1 1 4 7  
R2 2 5 8  
R3 3 6 9  
Enter 4 values for 2x2 matrix: 1 2 3 4  
  
2 x 2 Matrix:  
Col1 Col2  
Row1 1 2  
Row2 3 4  
> |
```

Environment Pane:

Object	Class	Attributes
m1	num	[1:5, 1:4] 1 5 9 13 17 2 6 10 14 18 ...
m2	num	[1:3, 1:3] 1 2 3 4 5 6 7 8 9
m3	num	[1:2, 1:2] 1 3 2 4
values	num	[1:4] 1 2 3 4